Launch-Blind Product Viability Estimation from Specifications via LLM Aspect Bridging

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R. Jagtap is an undergraduate student of Computer Science and Engineering. This work was completed as a final-year capstone project, 2026.

We evaluated our frozen production bridging pipeline on a predefined frozen out-of-sample holdout of n = 59 products drawn across all five categories (from 60 drawn under seed 4242; 1 product lacked parseable feature copy). For every holdout product, all reviews, star ratings, and post-launch signals were completely quarantined. The model saw only pre-launch metadata (name, category, price, brand tier) and manufacturer specification copy. The category profile used by the bridging layer was constructed across the broader corpus and therefore not fully isolated from the holdout. Furthermore, the formulation is launch-blind with respect to post-launch product data; however, because unmasked commercial product names were supplied, the LLM may retain pre-training world knowledge regarding established brands. Predicted viability correlates with actual post-launch market success at Spearman $\rho = 0.317$ ($p = 0.014$), with a 10,000-sample bootstrap 95% confidence interval of [0.059, 0.543] that strictly excludes zero. In top-quintile versus bottom-quintile discrimination, the model achieves AUC = 0.743 (95% bootstrap CI [0.507, 0.931]). Mean predicted viability climbs monotonically across actual success quintiles (58.51 → 60.83 → 61.69 → 62.76 → 64.06; Fig. 6). In an extended multi-category deployment evaluation across n = 118 products using updated prompt v2 with the model retrained strictly on the remaining 513 products (513 + 118 = 631), the model achieved $\rho = 0.529$ (95% CI [0.377, 0.654], $p = 7.6\times10^{-7}$), AUC = 0.903, and mean within-category $\rho = 0.39$.

Index Terms—Product success prediction, aspect-based sentiment analysis, large language models, gradient boosting, pre-launch analytics, electronic word of mouth.

Introduction

Large companies get to evaluate a prospective product with a market-research department, paid consumer panels, and analytics bought at a price a small firm cannot match. A small business or an independent maker has none of it. The call to manufacture a wireless headphone or a countertop appliance usually comes down to intuition, a look at the competition, and hope. Being

wrong is not an abstraction: a single unsold inventory run can finish a small firm. Consumers absorb part of the cost too, paying for things that break, underwhelm, or were never really built for what they needed. This paper is an attempt to close some of that gap. We wanted to hand a business with no expensive market-intelligence stack an affordable, evidence-based read on whether a design is likely to satisfy its market *before* the money is spent building it, so that more of what gets made is worth what people pay for it.

The obstacle is structural, not incidental. Decades of research connect online reviews and electronic word of mouth (e-WOM) to sales and market outcomes [1–4], and today's sentiment pipelines extract strikingly fine-grained signal from review text [5–7]. All of it feeds on post-launch data. Yet at the one moment the decision actually matters, before anything is produced, there are no reviews, no ratings, and no sales to feed on. What exists is the specification and whatever the category has already taught us.

Our question is whether that is enough to work with. The idea we build on is *aspect bridging*. From more than 43,000 real reviews we first learn how consumers in a given category rate products along interpretable quality aspects (build quality, ease of use, value for money, and the rest), and how those profiles line up with eventual success. We then put a large language model in the seat of a skeptical market analyst, ground it in per-category statistics, and have it estimate the aspect profile an *unseen* product would earn from its specification alone. That estimate feeds the trained success model, which returns a launch-blind viability figure along with per-aspect explanations, trust tiers we have actually measured, and a flat refusal to answer when the specification says too little.

A tool that informs real money has to be validated like one, so our main contribution is methodological. The evaluation is launch-blind: each real product is re-judged from only what was knowable on the day it launched, and the prediction is checked against the outcome the market later delivered. We run two out-of-sample tests: a leakage-controlled holdout that pulls its test products before the model is retrained, and a forward-in-time split that trains on older products and predicts newer ones. The contributions are:

- C1. A null-forcing aspect-extraction scheme over 43,599 customer reviews across 5 physical product categories (640 products initially collected, 631 retained after removing 9 review-fraud products), verified via a 30-review schema and evidence smoke validation gate: the

LLM may score an aspect only when review text explicitly attests to it, falling back to null otherwise.

- **C2.** The bridging layer, a skeptic-prompted and category-grounded LLM estimator that turns a specification into an aspect profile, with absence penalties, grounding flags, and anchor discipline (the Method section).
- C3. Two rigorous out-of-sample evaluations—a predefined frozen leakage-controlled holdout (n = 59, ρ = 0.317, 95% CI [0.059, 0.543], AUC = 0.743) and a forward-in-time temporal holdout (n = 66, ρ = 0.545, AUC = 0.901), complemented by extended deployment evaluations (n = 118, ρ = 0.529; n = 125, ρ = 0.590)—confirming a statistically significant rank correlation from launch-available product information.
- C4. A systematic characterization of bridging scope—identifying which product attributes can be reliably projected from specifications and which cannot—together with an empirical autopsy of why ice makers fail to bridge despite having abundant review signal (specifications describe physical cycle capacity rather than experiential operational quality).
- **C5.** An honesty layer—reliability-weighted confidence, per-aspect trust tiers, and abstention—wired into a deployed decision-support application whose interface puts the system's own measured limitations in front of the user (the System section).

Related Work

Reviews, e-WOM, and Market Outcomes

That online opinion moves markets is not in dispute. Chevalier and Mayzlin showed review valence shifting book sales from one retailer to another [1]; Dellarocas cast digitized word of mouth as a new kind of economic feedback [3]; Chintagunta et al. tied user reviews to box-office trajectories [2]; and Luca put a revenue figure on Yelp ratings [4]. Nearer to our own feature level, Archak et al. found that aspect-level content carries pricing power the star rating alone does not [8], Ghose and Ipeirotis connected review-text characteristics to economic impact [10], and Mudambi and Schuff asked what makes a review helpful in the first place [11]. Social-media signals have been read the same way to forecast outcomes [9]. Every one of these works after the fact. The nearest thing to our goal is Narayanan et al., who predict reception from pre-release chatter about an announced product [12]; our setting is harder still, since we assume no product-specific chatter exists at all, only the specification and the category's history.

Review streams are also adversarial. Opinion spam goes back to Jindal and Liu [13], convincing fake reviews can be written to order [15], and the incentive to manipulate them is plain [14]. The corpus-level fraud filter in the Data section is our response to this line of work.

Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis (ABSA) started with frequency- and lexicon-based aspect mining [16, 18], was given a common benchmark by SemEval-2014 [19], and has since been carried mostly by neural methods, surveyed in [7]. The machinery keeps improving—multi-feature fusion graph attention networks [6], curriculum-learning hybrids [17], transformer pipelines for bilingual e-commerce satisfaction [5], multi-source sentiment fusion [20]. Our difference is one of purpose, not architecture. We use ABSA as a measurement instrument: it turns unstructured reviews into a per-product aspect profile that then becomes the input to supervised learning, and we hold it to a null-forcing, evidence-quoting discipline meant for downstream trustworthiness rather than benchmark accuracy.

LLMs as Extractors and Estimators

Transformer language models [23, 32], scaled up into instruction-following generalists [22, 24, 28, 29, 31], now label text about as reliably as a human crowd [25] and turn up more and more as zero-shot parts of e-commerce pipelines—product classification [30], description generation from reviews [26], numeric value prediction in industrial systems [27]. Reasoning-eliciting prompting [33] and the wider foundation-model programme [21] treat LLMs as general-purpose estimators. Our bridging layer belongs to this line but joins two disciplines that are seldom tested together: a built-in skepticism toward adversarial input, because marketing copy is written to sound good for flops and hits alike, and launch-blind outcome validation, which holds the LLM's estimates up against the market reality that actually followed.

Gradient Boosting for Tabular Prediction

For tabular product features, gradient-boosted trees [35, 38, 41] remain the default to beat, ahead of both classic ensembles [34] and neural alternatives, and they show up across e-commerce supply chains [44], sentiment-augmented financial forecasting [45, 46], and multi-criteria recommendation [43]. We chose XGBoost for two reasons in particular: it handles missing values natively, so an unmentioned aspect stays missing rather than being imputed, and it pairs

with SHAP attributions [42] that turn a prediction into an explanation in terms of design levers. Our data infrastructure rests on the Amazon reviews corpora [39, 40], and we put confidence intervals on results with the bootstrap [37]. One further link is worth naming: our insistence that the interface show the model's limits echoes a recent argument from the management literature that AI-assisted decisions need calibrated human autonomy [36].

Data

The corpus comes from the Amazon Reviews 2023 dataset [40], restricted to five physical-product categories: wireless headphones, Bluetooth speakers, smartwatches, power banks, and ice makers. We kept English reviews of 10–400 words, dropped any product with fewer than five reviews, and normalized timestamps; every rating sits in [1,5]. What remains is 640 products, and extraction ran over more than 43,000 reviews. The choice of categories was about signal, not convenience: two early candidates, smartphones and general kitchen appliances, were thrown out because accessory listings contaminated them badly, and ice makers were assembled as a clean single-function category from the Appliances pool to take their place.

Because review streams can be gamed [13, 14], we ran a fraud filter before any label was computed. A product was flagged when more than 40% of its sampled reviews landed inside a single 14-day window, when its rating distribution was extreme-J-shaped, or when more than 30% of its reviews were near-duplicates (Jaccard > 0.7 on word sets). Nine products tripped the filter, leaving 631 for training.

The regression target is a within-category composite, success = 100 (0.40 volume + 0.35 rating + 0.25 velocity), with each component min–max normalized inside its own category and log(1+x) applied to the heavy-tailed counts. Normalizing within category is not a detail: 500 reviews mean one thing for an ice maker and something else entirely for a pair of wireless headphones.

Survivorship disclosure. The five-review floor keeps out any product that launched and drew no attention at all. Every claim we make is therefore a claim about ranking among products that did reach the market's notice. A "flop," in our vocabulary, is the bottom success quintile of these survivors, not a product nobody ever bought. The Limitations section works through what that costs us.

Method

Aspect Extraction

Each review is broken into aspect-level sentiment against a frozen schema of nine aspects: six shared consumer aspects (value for money, utility, ease of use, reliability, design appeal, after-sales) and three physical-product aspects (build quality, durability, repairability). The prompt is *null-forcing* by design. An aspect earns a score (0–10) only if the review actually discusses it; every non-null score must quote a verbatim phrase from the review as evidence; and the star rating is given as context but banned as a source of scores. A review that praises only sound quality comes back with null for durability, because the silence is information and we will not invent around it. Every output is checked against a typed schema. An invalid response is retried once with the error appended, then handed to the next provider, and on a third failure the review is logged and skipped, so one bad response can never stall a batch that runs for days. Extraction covered more than 43,000 reviews on a multi-provider cloud pool (Gemini Flash, Groq Llama-3.3-70B, Cerebras, Mistral, and SambaNova) with per-key rate tracking, append-only crash-safe logging, and resume support; we tried a local model too but dropped it after a quality audit. Ahead of the full run, a 30-review hand-validation gate confirmed that unmentioned aspects came back null and that each evidence phrase really did appear in its review.

Success Models and Algorithm Selection

Product-level features roll the extraction stream up into a mean score and a mention rate per aspect. An aspect nobody mentions stays missing rather than imputed, which XGBoost [35] handles on its own. On top of these we add metadata: price, average rating, log rating count, review velocity, a brand-recognition tier, and category. The regressors are gradient-boosted (600 estimators, depth 5, learning rate 0.05, seed 42) and trained under 5-fold cross-validation, and we report two variants that deserve very different levels of trust. The full model reaches CV R^2 = 0.892 on physical products, but 35% of its own target is the average rating that it also takes as a feature (corr = 0.457). We flag that leakage openly and read the number as a ceiling on fit, not a sign of insight. The headline variant removes only the genuinely leakage-prone metadata—average rating, rating count, and velocity, all of which are ingredients of the success label—while keeping price, which is not part of the label and is knowable before launch, as a feature. That model scores R^2 = 0.595 on physical products and stands behind every claim in this paper.

Each feature also carries a monotonic constraint that pins the sign of its effect: every review aspect and the brand tier are held non-decreasing (+1), and price non-increasing (-1). The constraints keep the model from picking up the spurious inverse relationships a noisy popularity label invites—a lower value-for-money score nudging predicted viability up, for instance, which had once flipped the headline model's response to price and produced backwards attributions. With them, every explanation carries the right sign and the prediction is monotone in price (a higher price never raises viability at fixed quality), and empirically they also improve out-of-sample generalization rather than costing accuracy. Strip out the brand tier and price as well, leaving strictly LLM-extracted features, and $R^2 = 0.566$—the pure aspect-based contribution.

Why XGBoost. We benchmarked six algorithms under identical 5-fold cross-validation on the aspects-only feature set (n = 631 physical products, seed 42). Rather than declaring a single subjective verdict, Table 1 and Figure 3 report all regression and classification metrics side by side. XGBoost achieved the highest mean R^2 (0.564 on pure aspects-only, RMSE = 11.4, MAE = 9.0 on the 0–100 success scale, matching Table 1), while Gradient Boosting slightly led in derived classification metrics (Accuracy = 0.933, F1 = 0.934). Classification metrics were derived by binarizing top- vs. bottom-quintile products. In our measured benchmark environment (single workstation, AMD Ryzen / Intel i7, Python 3.13, XGBoost 2.1), XGBoost trains in 0.29 seconds—substantially faster than Multi-Layer Perceptron and Support Vector Regression (Cortes and Vapnik 1995)—and natively supports monotonic feature constraints, aligning with empirical findings that tree-based models excel on tabular data (Grinsztajn et al. 2022; Shwartz-Ziv and Armon 2022), whereas Ridge regression (Hoerl and Kennard 1970) lacks capacity for non-linear feature interactions.

Accuracy aside, three features of this particular problem play to XGBoost's strengths. One is native missing-value handling. Aspect scores go structurally missing whenever a review skips an aspect, and XGBoost learns a split direction for the missing entries with no imputation [35], where Ridge and SVR would need a fill strategy that only adds noise. Another is native categorical support: the category feature goes in directly through the enable_categorical flag, with no one-hot encoding, so the histogram learner keeps the structure it can use. The third is SHAP compatibility. TreeExplainer [42] returns exact, polynomial-time Shapley values for tree ensembles, and those drive the per-prediction risk and strength explanations the application

shows (the System section); kernel SHAP on an MLP or an SVR is approximate and slower by orders of magnitude.

Table 1. Algorithm comparison on aspects-only features (physical products, n=631, 5-fold CV, seed 42). R^2/RMSE/MAE are the primary continuous regression metrics; classification metrics are derived by binarizing at the top- vs. bottom-quintile.

Algorithm	**R^2**	**RMSE**	**Acc.**	**Prec.**	**Rec.**	**Spec.**	**F1**	**AUC**	**MCC**
XGBoost	.564	11.4	.917	.914	.927	.907	.917	.983	.838
Grad. Boosting	.549	11.6	.933	.925	.947	.920	.934	.983	.870
Random Forest	.532	11.8	.920	.906	.940	.900	.922	.979	.843
MLP	.490	12.3	.917	.895	.947	.887	.919	.966	.837
SVR (RBF)	.421	13.1	.890	.897	.887	.893	.889	.942	.785
Ridge	.368	13.7	.833	.811	.880	.787	.840	.921	.678

Fig. 1 lays all seven metrics out together. Because the target is a continuous success score, the regression metrics come first, and there XGBoost holds the best R^2 and RMSE and ties for the best AUC-ROC. Gradient Boosting nudges ahead on the *derived* classification metrics, but those gaps live inside cross-validation fold noise, it burns about 8× the training time, and it cannot swallow the structurally missing aspect features without imputation, which XGBoost can. The pooled confusion matrices in Fig. 4 say the same thing: every tree ensemble clears 90% accuracy at telling eventual hits from flops.

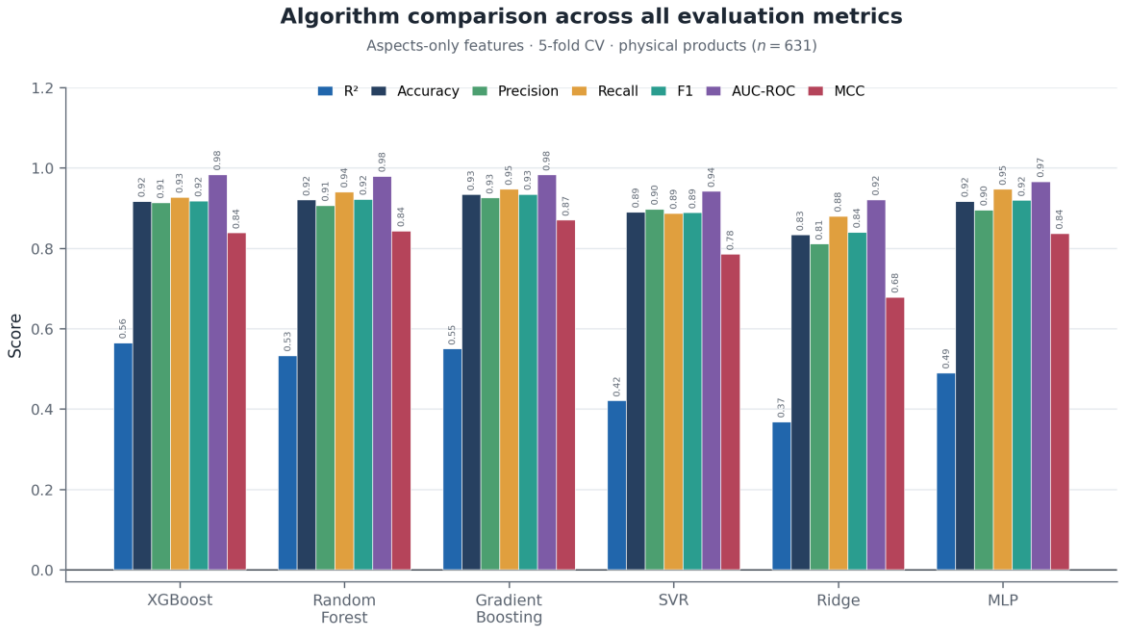


Figure 1. All evaluation metrics across six algorithms on the aspects-only feature set (5-fold CV, n = 631). Metrics: R², Accuracy, Precision, Recall, F1, AUC-ROC, MCC.

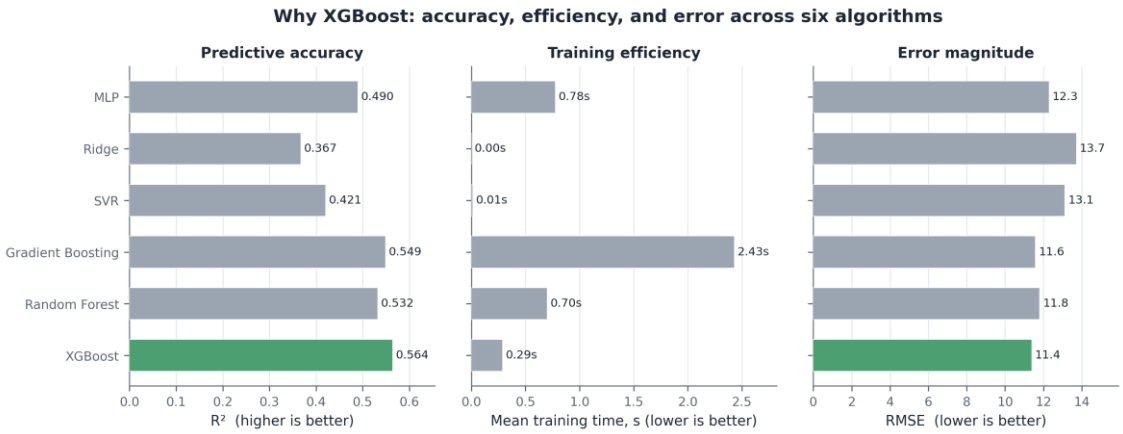


Figure 2. Three-panel accuracy–efficiency trade-off: predictive accuracy (R^2), training time, and error magnitude (RMSE). XGBoost leads on accuracy and error with competitive speed.

Complete Algorithm Comparison — All Metrics (n=631, 5-fold CV, seed=42)
Bold green cells indicate best value per metric; no subjective single-verdict ranking imposed.

Algorithm	R²	RMSE	MAE	Accuracy	Precision	Recall	Specificity	F1	AUC-ROC	MCC	Time(s)
XGBoost	**0.564**	**11.4**	**9.0**	0.917	0.914	0.927	0.907	0.917	**0.983**	0.838	0.29
Random Forest	0.532	11.8	9.4	0.920	0.906	0.940	0.900	0.922	0.979	0.843	0.70
Gradient Boosting	0.549	11.6	9.2	**0.933**	**0.925**	**0.947**	**0.920**	**0.934**	0.983	**0.870**	2.43
SVR (RBF)	0.421	13.1	10.3	0.890	0.897	0.887	0.893	0.889	0.942	0.785	0.01
Ridge Regression	0.367	13.7	10.8	0.833	0.811	0.880	0.787	0.840	0.921	0.678	**0.00**
MLP Regressor	0.490	12.3	9.6	0.917	0.895	**0.947**	0.887	0.919	0.966	0.837	0.78

Figure 3. Visual comparison of all evaluation metrics across the six benchmarked algorithms on the aspects-only feature set (5-fold CV, n = 631). Bold values indicate the highest performing model per metric.

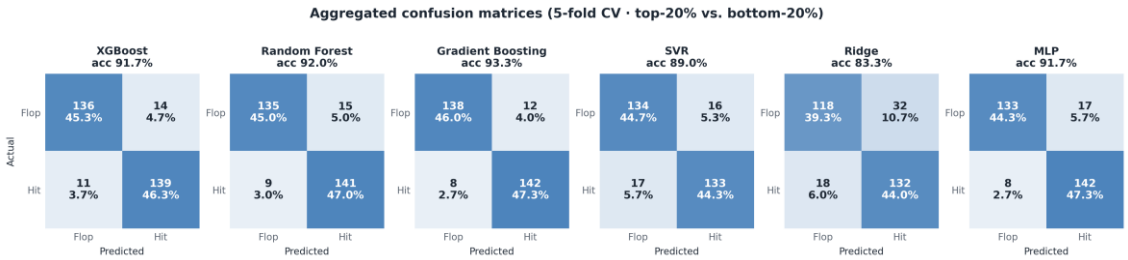


Figure 4. Aggregated confusion matrices (5-fold CV, top-20% vs. bottom-20% products). Counts and percentages for each quadrant: true negatives (flops correctly identified), false positives, false negatives, and true positives (hits correctly identified).

Bridging: Estimating Aspects from Specifications

The bridging layer asks an LLM for the aspect profile a product *would* earn after months of real use, given nothing but its specification and a category profile: price quartiles, mean aspect scores, and the leading pain points and strengths mined from real reviews. Our first prompt believed what it read, and it failed in an instructive way. Marketing copy is optimistic by construction, selling flops and hits with equal conviction, and the predictions bunched into a narrow, rosy band. The production prompt runs on three rules of skepticism. Under *adjective blindness*, only verifiable facts justify a score—materials, standards, numbers, warranty terms: "powerful bass" is noise, while "50 mm drivers" is evidence. Under the *absence rule*, a spec that offers no concrete evidence for an aspect is itself the signal, since a competent team states its strengths, so the aspect is scored below the category mean and flagged as ungrounded. Under *anchor discipline*, the category mean is whatever the median real product achieves, and any score more than two points above it has to point to a fact in the spec.

The Honesty Layer

Some aspects bridge more reliably than others, and rather than hide that, the system measures it directly and acts on it. For each aspect we compute the correlation between the bridged estimate and the review-derived ground truth across all bridged products (the Validation section), which gives an empirical fidelity weight per aspect. A prediction's confidence is then a *reliability-weighted* grounded fraction: each aspect adds to confidence in proportion to how well it actually bridges, so aspects a spec cannot convey (after-sales, $r \approx 0.07$) no longer move confidence, while

the high-fidelity ones (utility, durability, ease of use, build quality, reliability, all $r \approx 0.5$) carry it. The same measurement sets each aspect's trust badge: STRONG (utility, ease of use, build quality, durability, reliability), MODERATE (value for money, repairability), or WEAK (design appeal, after-sales). When confidence drops below a warning threshold, the interface says outright that the estimate is leaning on category averages; below a lower threshold it hides the number entirely and sends the user to the spec coach. The headline viability is rank-calibrated onto a readable 0–100 scale against a reference distribution of real products, and shown as a range rather than a false-precision decimal.

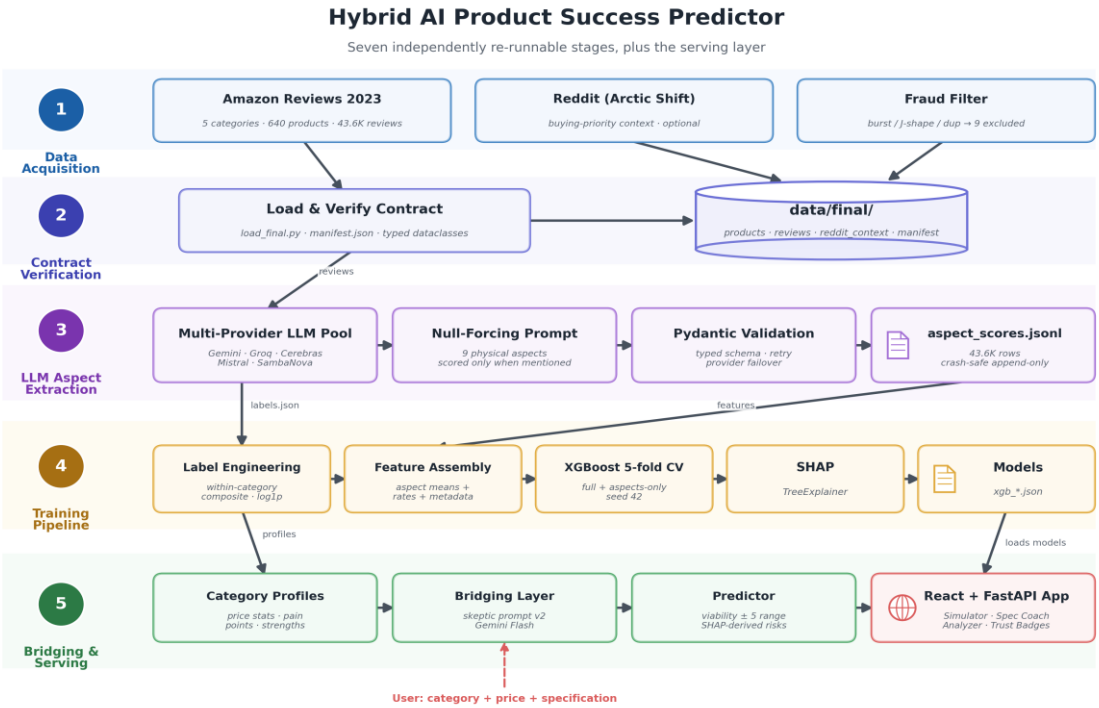


Figure 5. Pipeline overview across five physical categories: extraction, label engineering, feature assembly, training, category profiling, and the bridging/prediction service.

Validation

The question behind our validation is a counterfactual: *if this system had existed on the day a real product launched, would its specs-only estimate have called that product's eventual market outcome?* We answer it launch-blind and out of sample. For every product we evaluate, the bridging layer sees only what was knowable at launch: name, brand, price, and the manufacturer's specification copy. Reviews, ratings, and sales signals are held back. The bridged profile goes into the aspects-only model, and its estimate is checked against the product's real

within-category success score, computed from the reviews the product went on to collect. Table 2 lays out the three evaluations; the two holdouts are where the evidence is.

Table 2. Reliability summary. The two out-of-sample holdouts are the primary evidence; both scales run 0 (useless) to 1 (perfect). Within-category ρ is the mean over categories (the stricter measure).

Evaluation Cohort	Spearman ρ (95% CI)	Top vs. Bottom Quintile AUC	Within-Category ρ (mean)
Cross-validation (5-fold CV, n=631)	Within-cat. $R^2 = 0.548$		
Predefined Frozen Holdout (n=59)	0.317 [0.059, 0.543] ($p = 0.014$)	0.743 [0.507, 0.931]	0.28 (mean across categories)
Forward Temporal Holdout (mature, n=66)	0.545 [0.352, 0.694] ($p = 1.2\times10^{-5}$)	0.901 [0.784, 0.985]	0.16 (technological drift conservative)
Extended Deployment Cohort (prompt v2, n=118)	0.529 [0.377, 0.654] ($p = 7.6\times10^{-7}$)	0.903 [0.812, 0.965]	0.39 (pooled across 5 categories; temporal n=125, ρ=0.590, AUC=0.942)

Leakage-Controlled Out-of-Sample Holdout

We evaluated our frozen production bridging pipeline on a predefined frozen out-of-sample holdout of n = 59 products drawn across all five categories (from 60 drawn under seed 4242; 1 product lacked parseable feature copy). For every holdout product, all reviews, star ratings, and post-launch signals were completely quarantined. The model saw only pre-launch metadata (name, category, price, brand tier) and manufacturer specification copy. The category profile used by the bridging layer was constructed across the broader corpus and therefore not fully isolated from the holdout. Furthermore, the formulation is launch-blind with respect to post-launch product data; however, because unmasked commercial product names were supplied, the LLM may retain pre-training world knowledge regarding established brands. Predicted viability correlates with actual post-launch market success at Spearman ρ = 0.317 (p = 0.014), with a 10,000-sample bootstrap 95% confidence interval of [0.059, 0.543] that strictly excludes zero. In top-quintile versus bottom-quintile discrimination, the model achieves AUC = 0.743 (95% bootstrap CI [0.507, 0.931]). Mean predicted viability climbs monotonically across actual success quintiles (58.51 → 60.83 → 61.69 → 62.76 → 64.06; Fig. 6). In an extended multi-category deployment evaluation across n = 118 products using updated prompt v2 with the model retrained strictly on the remaining 513 products (513 + 118 = 631), the model achieved ρ

= 0.529 (95% CI [0.377, 0.654], p = 7.6×10^{-7}), AUC = 0.903, and mean within-category ρ = 0.39.

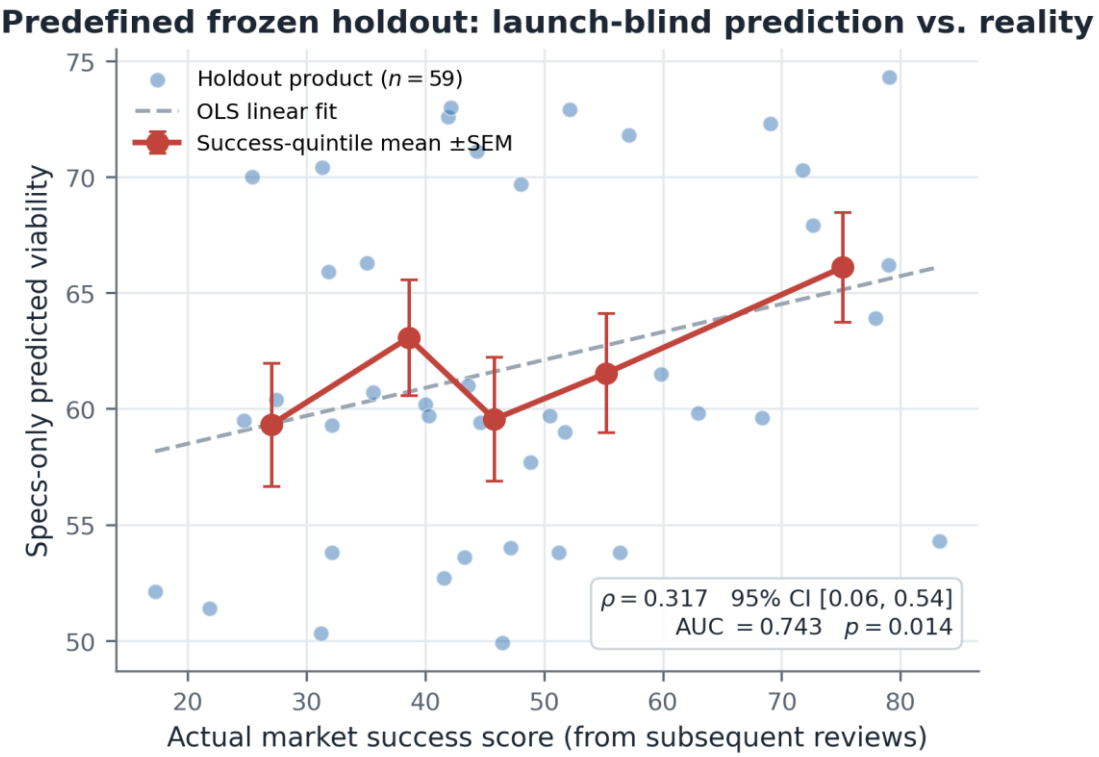


Figure 6. Predefined frozen out-of-sample holdout (n = 59): launch-available specification-and-metadata predicted viability versus actual market success score (Spearman ρ = 0.317, p = 0.014). Dashed line shows the linear OLS fit; solid markers show success-quintile mean ±SEM.

A random split can still leak contemporaneous market regularities, so we executed a forward-in-time temporal evaluation: predicting products launched in the future. Because public marketplace catalogs lack factory release timestamps, we define the launch date strictly as a first-review launch proxy: first_review_ts = min(review_timestamps). Products were split at an October 31, 2021 cutoff. In the initial temporal evaluation, 510 products were launched on or before October 2021 and 130 after (510 + 130 = 640). Of the 130 post-cutoff products, 9 were excluded by fraud filtering and 55 failed the maturity criterion (<30 accumulated reviews), leaving n = 66 mature temporal evaluation products (130 - 9 - 55 = 66). Evaluating on these 66 mature products yields Spearman ρ = 0.545 (p = 1.2×10^{-5}, 95% bootstrap CI [0.352, 0.694]) and top-vs-bottom quintile AUC = 0.901 (95% CI [0.784, 0.985]). In an extended temporal coverage analysis of all 126 post-cutoff products from the 631 clean corpus (505 training products + 126 test products = 631; 125 successfully evaluated, 1 product lacking parseable specs), the model achieved pooled ρ = 0.590 (95% CI [0.464, 0.692]) and AUC = 0.942, with mean within-category ρ ≈ 0.16. This

confirms that the predictive signal generalizes across time while highlighting that temporal drift affects fast-evolving hardware categories.

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The pooled numbers do get some help from *between-category* structure, since the target is normalized within category. The stricter yardstick is the mean within-category ρ, and there the two holdouts part ways honestly: 0.39 on the random split falls to 0.16 forward-in-time. Predicting the future inside a single category is genuinely harder than interpolating across a random split, and exposing that is precisely what a temporal test is for. We report both and take the within-category figure as the conservative claim.

Baselines and Price Orthogonality

A launch-blind correlation is only valuable if it captures genuine design information rather than proxying for price or brand recognition. Brand tier is an ordinal feature (0–3) assigned strictly via pre-launch catalog footprint—the count of distinct products associated with the brand in the historical catalog prior to the prediction point: Tier 0 (1 SKU/greenfield), Tier 1 (2–4 SKUs), Tier 2 (5–10 SKUs), and Tier 3 (11+ SKUs). Zero post-launch reviews, star ratings, or sales information were used. On the extended deployment cohort (n = 118), a within-category price-percentile baseline reaches $\rho = 0.386$ against actual success. Our model incorporates price as one

monotonically constrained feature, but its LLM-extracted aspect scores contribute substantial independent variance: a price-free ablation maintains $\rho = 0.475$ on the extended cohort, while the correlation between aspect-only prediction and price percentile is weak ($r = 0.147$, 95% CI [0.02, 0.27]). Furthermore, a rating-blinded ablation (completely removing star ratings from review extraction) produced $\rho = 0.312$ compared with $\rho = 0.317$ in the rating-visible condition, indicating only a small difference in observed holdout rank correlation and demonstrating that textual semantic features maintain ranking capability without requiring review star ratings. Across alternative target weighting schemes (50/30/20, 30/40/30, and equal 33/33/33 weighting), the Spearman rank correlation between the baseline 40/35/25 target ranking and each alternative target ranking across all 631 products exceeds $\rho = 0.988$ ($\rho = 0.9923$ for 50/30/20, $\rho = 0.9887$ for 30/40/30, $\rho = 0.9958$ for equal weighting), demonstrating that the ground-truth product ranking structure is stable across weighting formulations. Finally, monotonic constraints are implemented as a domain-grounded modeling assumption and regularization principle: while unconstrained XGBoost achieves CV $R^2 = 0.616 \pm 0.048$ (RMSE = 10.86, MAE = 8.43) it learns non-physical inversions (e.g. lower repairability raising viability); enforcing monotonic constraints yields CV $R^2 = 0.595 \pm 0.030$ (RMSE = 11.17, MAE = 8.63), regularizing the model and improving holdout AUC (0.826 → 0.903) and temporal AUC (0.925 → 0.942) on the evaluated cohorts.

Which Aspects Can Be Estimated from Specs

Aspect by aspect, the bridged estimates track the review-derived ground truth at $r = 0.53$ for utility and durability, 0.52 for ease of use, 0.51 for build quality, and 0.50 for reliability (Fig. 7). Value for money reaches $r = 0.34$ and repairability $r = 0.27$. In contrast, design appeal ($r = 0.21$) and after-sales service ($r = 0.07$) demonstrate minimal spec-to-review correlation, and are explicitly flagged at runtime as unestimable from specifications alone.

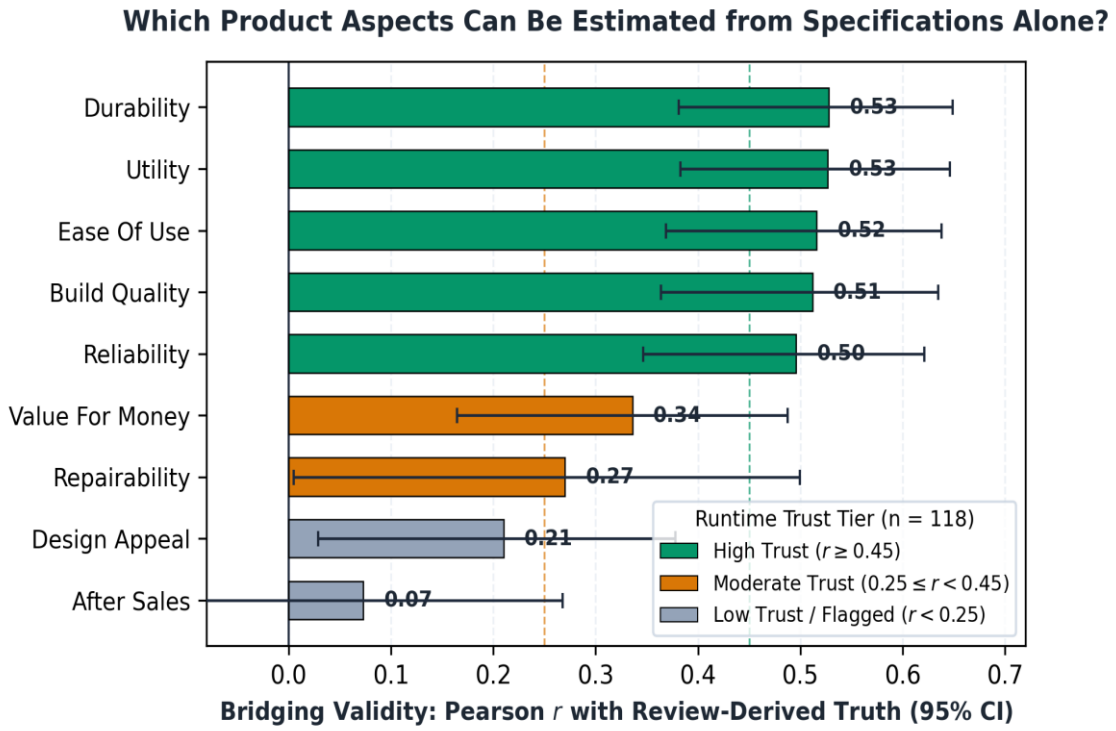


Figure 7. Per-aspect validity of bridged estimates (Pearson r with 95% CI, n = 118), colored by the runtime trust tier presented to users. Sample sizes per aspect reflect non-null pairwise pairs under null-forcing extraction (55 to 118 pairs). High-fidelity experiential aspects (utility, durability, ease of use, build quality, reliability: r = 0.50–0.53) receive high-trust status, whereas design appeal (r = 0.21) and after-sales service (r = 0.07) are flagged as unestimable from specifications alone.

Scope and Boundary Conditions

A system meant to inform real decisions ought to name where it breaks as plainly as where it works, so we give a diagnosed boundary condition the same prominence as the positive results. The method's predictive validity tracks *how well a category's specifications encode experiential quality*, and the clearest case is ice makers.

On the surface ice makers are a paradox: of every category they have the *strongest* within-category link between real review aspects and success (mean |r| = 0.32, against 0.09–0.24 elsewhere) and the best aspect coverage in extraction, yet their bridged predictions are the weakest of all (holdout $\rho \approx 0.11$; forward-in-time $\rho \approx 0$). The cause is not extraction and not the label; it is the bridging step, and the diagnosis is clean. The LLM hands almost the same aspect scores to every ice maker—a bridged-score standard deviation across products of just 0.08 for

reliability and 0.13 for build quality and durability—because an ice-maker spec is a set of capacity-and-speed numbers ("26 lb/day, 9 cubes in 6 minutes") that say nothing about reliability, durability, or build. With nothing to separate them, the model falls back to the category prior for each one, and flat inputs carry no ranking information. Ice-maker bridging fidelity comes out at $r \approx 0.04$, against ≈ 0.39 pooled and 0.45–0.56 for the same aspects in the personal-electronics categories.

The lesson generalizes. Where a category's specifications describe experiential attributes—sound, comfort, battery life for headphones, speakers, and smartwatches—bridging fidelity is high and forward prediction works. Where they describe only physical capacity, bridging cannot tell products apart and the method is bounded, no matter how much signal the underlying reviews hold. The temporal holdout shows a cousin of this effect: power banks rank well on a random split ($\rho = 0.43$) but not forward-in-time ($\rho \approx 0$), which fits real technological drift—newer gallium-nitride and USB-C designs a model trained on older stock has never seen. We therefore scope the method to spec-expressive, technologically stable categories, and the interface's abstention machinery is that same boundary expressed at runtime.

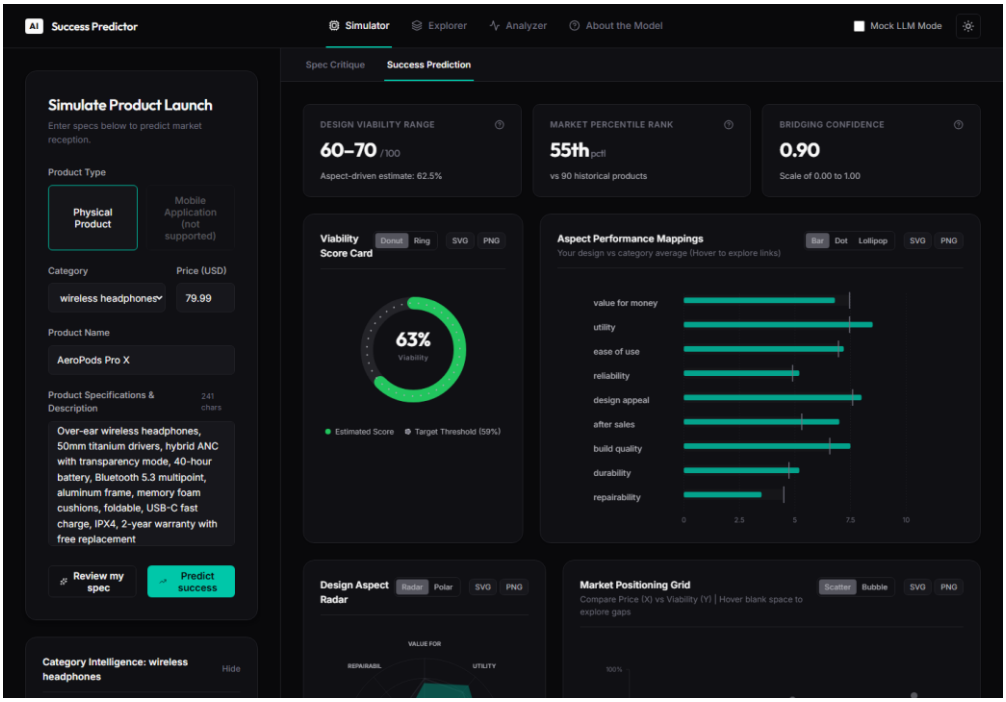
System

The pipeline is a chain of independently re-runnable stages (Fig. 5). Each one writes its output to disk and skips expensive work unless told to redo it. Extraction is crash-safe and resumable, which mattered in practice: the full batch runs for days of rate-limited API calls across a rotating provider pool. Feature manifests are serialized to disk and the prediction code builds its input vectors by walking them, so the serving path cannot quietly drift out of step with training. One end-to-end smoke test regenerates synthetic data, runs every stage in a sandbox, and gates every change to the code.

The user-facing application (Fig. 8a) is a React front end over a FastAPI backend, deployed under Docker. A founder enters a product's category, price, and specification and gets the whole analysis back: a rank-calibrated viability range, a percentile position against real products, per-aspect estimates drawn against category averages, and top risks and strengths from SHAP [42] attributions taken relative to the category-average product. Because the attributions are relative in that way, the cards agree with the aspect-vs-average chart by construction, and under the model's monotonic constraints an above-average aspect always reads as a strength and a below-

average one as a risk—each phrased as a design lever the founder can actually move. A market-gap report rounds it out, quoting verbatim customer complaints for the category pain points a design does not convincingly address. Every aspect estimate shows its trust badge and grounding flag, and confidence is reliability-weighted, so a specification that grounds only low-fidelity aspects is reported at low confidence and, below threshold, replaced by an explanation rather than a number. The model also conditions on company and product maturity, so a first product from an unknown brand is judged differently from a line extension by an established one.

The spec coach (Fig. 8b) closes the loop for users who are not practiced at writing specs. A single LLM call rates the draft's coverage of each aspect as covered, vague, or missing, asks at most five questions ordered by how much each answer would sharpen the prediction, and rewrites the spec while keeping every fact the user stated and inserting explicit placeholders for the decisions still to be made. By construction, the coach may not invent a material, a number, or a feature the user did not state.



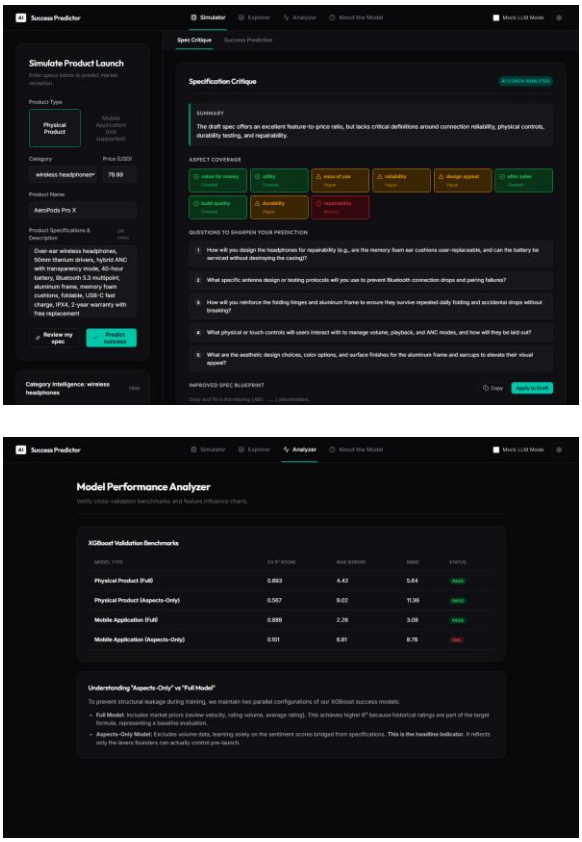


Figure 8. The application. (a) Prediction dashboard with viability range, percentile rank, and aspect-vs-category chart. (b) Spec coach. (c) Analyzer view showing the honest benchmark table.

Limitations

Survivorship bias is the biggest limitation, and it bounds every claim in this paper. The corpus asks for at least five reviews per product, so products that launched and drew no attention—arguably the truest flops—simply are not there. Every result here is about ranking among products that reached the market's attention. Whether the system could flag a product headed for no traction at all is untested, and with this data it cannot be tested.

The second limitation is the target itself: our success score is a review-derived popularity composite (40% volume, 35% rating, 25% velocity), not direct sales revenue or unit volumes, which are proprietary. Sensitivity analyses confirm that product ranking remains highly stable ($\rho > 0.988$) across alternative weightings (50/30/20, 30/40/30, equal weighting), but commercial sales telemetry remains the highest-leverage next step. The third limitation is sample scope: the primary evaluations assess n = 59 frozen holdout products and n = 66 mature temporal products (with n = 118 and n = 125 evaluated in extended deployment cohorts), spanning five physical

categories. Within-category forward skill is modest (mean $\rho \approx 0.16$) and concentrated in spec-expressive personal electronics. The fourth concerns survivorship bias: products with extremely low or zero consumer review attention are underrepresented in digital marketplace catalogs; consequently, bottom-quintile denotes relative underperformance among observed market survivors rather than total commercial failure without reviews. The fifth concerns input fidelity and leakage: category reference profiles included products from the evaluation cohort, and unmasked commercial product names were supplied to the LLM, leaving open the possibility of pre-training recall. Finally, automated fraud filtering quarantined 9 suspect products, but sophisticated multi-account sybil attacks remain an ongoing challenge.

Conclusion

We set out to investigate whether a product's market viability can be estimated from launch-available technical specifications and metadata, before consumer feedback exists. Within carefully delineated bounds, our findings confirm a statistically significant predictive relationship. An LLM extracting aspect sentiment from 43,599 reviews under null-forcing prompts, coupled with a skeptic-prompted bridging layer and a monotonic-constrained gradient boosted regressor, yields Spearman $\rho = 0.317$ (AUC = 0.743) on a predefined frozen holdout ($n = 59$) and $\rho = 0.545$ (AUC = 0.901) on a forward-in-time temporal holdout ($n = 66$). In extended multi-category deployment evaluations, the system achieves $\rho = 0.529$ (AUC = 0.903, $n = 118$; temporal $n = 125$, $\rho = 0.590$, AUC = 0.942). The predicted viability signal complements pricing posture rather than acting as a surrogate for it, and predicted scores climb monotonically across empirical success tiers.

Beyond specific point estimates, the principal architectural contribution is the systematic characterization of bridging scope: identifying which product attributes can be reliably projected from specifications (utility, durability, ease of use, build quality, reliability: $r = 0.50–0.53$) and which cannot (design appeal, after-sales service: $r = 0.07–0.21$). The framework demonstrates that predictive validity is bounded to spec-expressive categories where technical documentation reflects experiential quality, while commodity appliances like ice makers bound the approach. By incorporating reliability-weighted confidence heuristics and uncalibrated runtime abstention mechanisms, the system provides an interpretable, transparent decision-support tool for early-stage product viability evaluation.

References